

Nickel-catalyzed C(sp²)-C(sp²) Cross Coupling Reactions of Sulfur-Functionalities and Grignard Reagents

Chika I. Someya · Maik Weidauer ·
Stephan Enthaler

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Abstract In the present study, the nickel-catalyzed carbon carbon bond formation of a range of sulfur containing substrates with Grignard reagents via desulfurization has been explored. After investigation of different reaction parameters with a well-defined nickel complex an excellent and easy-accessible pre-catalyst was found. The obtained system was capable to convert a broad scope of substrates under mild reaction conditions.

Keywords Catalysis · Cross coupling · Nickel · Sulfur compounds · Grignard reagents

1 Introduction

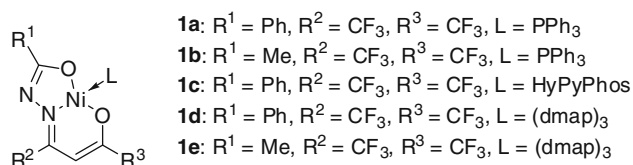
During the last decades nickel-catalyzed carbon–carbon bond formations have been extensively studied and nowadays it is a methodology of choice with a broad applicability in the synthesis of fine chemicals, pharmaceuticals, agricultural chemicals and natural products [1, 2]. Commonly, for creating a carbon–carbon bond organic halides were reacted with organometallic reagents in the presence of nickel catalysts [3–6]. Contrary to the well explored halides as leaving group the application of other functions (e.g., C–OC, C–CN, C–N, C–S) has been less explored. Interestingly, such functions can be valuable for the creation of complex molecules since a different reactivity compared to halides is feasible. In this regard, the conversion of carbon–sulfur bonds to carbon–carbon bonds

can be a useful tool [7–10]. In 1979 the nickel based cross-coupling of thioethers or thiols and Grignard reagents have been reported by Wenkert and Takei et al. applying [NiCl₂(PPh₃)₂] as pre-catalyst [11–16]. Afterwards the methodology was advanced for instance to the transformation of sulfoxides, sulfones, sulfoximines, sulfonium salts or thiol esters with Grignard or organometallic reagents [7–10, 17–46]. More recently, the group of Knochel demonstrated the value of the concept in the cross coupling reactions of various thio derivatives and functionalized aryl-, heteroaryl-, alkyl-, and benzylic zinc reagents [47–51]. Recently we have demonstrated the usefulness of complexes **1** in nickel-catalyzed C–C cross coupling reactions of organic halides with organometallic zinc reagents (Scheme 1) [52–54]. Moreover, excellent reactivity was observed in the hydrodehalogenation of aryl and alkyl halides and in the hydrodecyanation of aryl cyanides applying *tert*-butylmagnesium chloride as reagent [55, 56]. On the other hand complexes **1** allow the hydrodesulfurization of organic compounds by cleavage of carbon sulfur bonds [55, 56]. In this regard, we report herein our studies on the nickel-catalyzed carbon carbon bond formations via desulfurization reactions.

2 Results and Discussion

Initially, the cross coupling of thioanisole (**2**) with 4-anisylmagnesium bromide (**3**) in THF was studied as a model reaction to explore appropriate conditions and to examine the influence of various reaction parameters (Table 1). As expected when performing the reaction in the absence of nickel complexes no product formation was monitored (Table 1, entry 1). However, when applying 2.5 mol% of **1a** 72 % of the cross coupling product **4** was obtained after

C. I. Someya · M. Weidauer · S. Enthaler (✉)
Cluster of Excellence “Unifying Concepts in Catalysis”,
Department of Chemistry, Technische Universität Berlin,
Straße des 17. Juni 115/C2, 10623 Berlin, Germany
e-mail: stephan.enthaler@tu-berlin.de



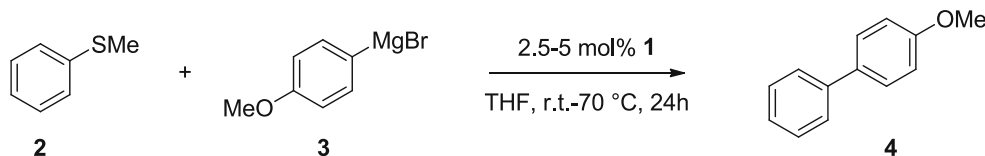
Scheme 1 Selection of nickel complexes applied in cross coupling reactions

24 h at 70 °C (Table 1, entry 2). Noteworthy, as side reaction the homo coupling of **3** and the homo coupling of **2** (<7 %) were observed. The yield of the desired product was increased to 77 % by increasing the catalyst loading to 5.0 mol % (Table 1, entry 3). Interestingly, the nickel catalyst demonstrated activity at room temperature (Table 1, entry 5). Moreover, the influence of the ligand structure of the nickel complex was studied (Table 1, entries 10–15). A better performance was observed for square planar phosphane complexes (**1a–1c**), while for the octahedral 4-dimethylaminopyridine complexes (**1d–1e**) a reduced yield was monitored. In more detail, within the phosphane systems the triphenylphosphane ligand was

favoured. Recently, the group of Miller reported on the increase of reactivity by addition of LiO^tBu to Grignard reagents [57–60]. Interestingly, the addition of 1.5 equiv. of LiO^tBu to the reaction mixture lead to full conversion after 1 h, probably by de-aggregation of organomagnesium clusters or increasing the basicity of the Grignard reagent (Table 1, entry 6) [61]. Noteworthy, in the absence of the nickel complex no product formation was observed, excluding a LiO^tBu mediated reaction pathway. In general, the realized catalyst performance is to some extent improved in comparison to the earlier system established by Wenkert and coworkers, while 10 mol% of $[\text{NiCl}_2(\text{PPh}_3)_2]$ resulted in the formation of less than 80 % yield of the desired products at higher temperature (80 °C) [11–16].

Once the optimized reaction conditions were established (Table 1, entry 7) the scope and limitations of the nickel-catalyzed cross coupling of substrates containing thio-functionalities and Grignard reagents were investigated. Moreover, the reaction outcome was studied in the presence and absence of stoichiometric amounts of LiO^tBu . First different *para*-substituted thioanisoles were reacted with 4-anisylmagnesium bromide (**3**). An excellent yield was

Table 1 Nickel-catalyzed cross coupling of thioanisole (**2**) and 4-anisylmagnesium bromide (**3**) – optimization of the reaction conditions



Entry ^a	Pre-catalyst [mol%]	T [°C]	T [h]	Sel. [%] ^b	Yield [%] ^c
1	–	70	24	<1	<1
2	1a (2.5)	70	24	95	72
3	1a (5.0)	70	24	94	77
4	1a (5.0)	70	24	>99	81 ^d
5	1a (5.0)	r.t.	24	95	63
6	1a (5.0)	70	24	>99	>99 ^e
7	1a (5.0)	70	24	93	87 ^f
8	1a (5.0)	70	1	>99	>99 ^e
9	–	70	1	<1	<1 ^e
10	1b (5.0)	70	24	95	73
11	1b (5.0)	70	24	>99	>99 ^e
12	1b (5.0)	70	1	>99	>99 ^e
13	1c (5.0)	70	24	99	72
14	1d (5.0)	70	24	96	67
15	1e (5.0)	70	24	>99	39

^a Reaction conditions: **2** (0.2–0.6 mmol), pre-catalyst **1** (2.5–5 mol%), **3** (1.5 equiv., 1.0 M in THF), THF (2.0 mL), r.t.-70 °C, 1–24 h

^b Selectivity. As side product 1,1'-biphenyl was observed

^c Determined by GC–MS applying *n*-dodecane as internal standard

^d 2.0 equiv. **3**

^e 1.5 equiv. LiO^tBu

^f (5.0 mmol), pre-catalyst **1** (5 mol %), **3** (1.5 equiv., 1.0 M in THF), THF (7.5 mL), 1.5 equiv. LiO^tBu , isolated yield 66 %

Table 2 Nickel-catalyzed cross coupling of thio-functionalities and 4-anisylmagnesium bromide (**3**)—scope and limitations

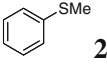
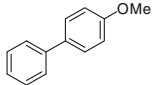
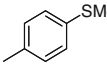
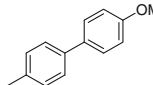
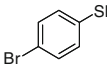
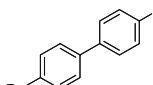
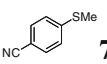
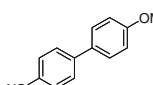
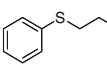
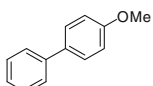
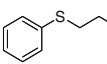
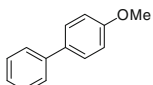
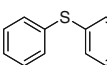
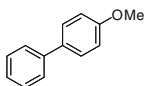
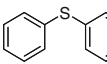
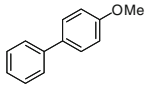
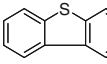
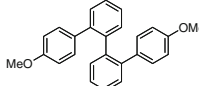
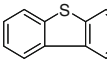
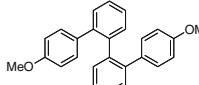
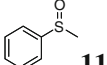
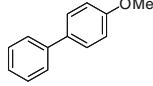
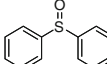
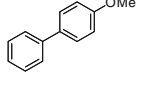
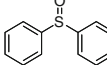
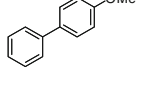
$ \begin{array}{c} \text{R}^1\text{-S-R}^2 + \text{MeO-C}_6\text{H}_4\text{-MgBr} \xrightarrow[\text{THF, 70 } ^\circ\text{C, 24h}]{\text{5 mol\% } \mathbf{1a}, \text{ 1.5 equiv. LiO}^t\text{Bu}} \text{MeO-C}_6\text{H}_4\text{-R}^1 \\ \mathbf{2,5-15} \qquad \qquad \qquad \mathbf{3} \qquad \qquad \qquad \mathbf{4,5a-7a,10a,14a} \end{array} $				
Entry ^a	Substrate	Product	Sel. [%] ^b	Yield [%] ^c
1	 2		>99 (94)	4 : >99 (77)
2	 5		>99 (>99)	5a : 96 (62)
3	 6		<1 (<1)	6a : <1 (<1) ^d
4	 7		>99 (<1)	7a : 21 (<1) ^c
5	 8		>99 (>99)	4 : 43 (51)
6	 8		>99 (>99)	4 : 84 (85) ^f
7	 9		92 (94)	4 : 66 (73)
8	 9		97	4 : 69 ^g
9	 10		>99 (>99)	10a : 30 (36)
10	 10		>99 (>99)	10a : 70 (37) ^f
11	 11		>99 (>99)	4 : 50 (35)
12	 12		>99 (>99)	4 : 78 (81)
13	 12		>99	4 : 42 ^g

Table 2 continued

Entry ^a	Substrate	Product	Sel. [%] ^b	Yield [%] ^c
14			>99 (>99)	5a : 70 (65)
15			>99	5a : 50 ^g
16			<1 (70)	14a : <1 (21)
17			>99 (73)	4 : 80 (73)
18			>99	4 : 68 ^g

^a Reaction conditions: substrate (0.3 mmol), pre-catalyst **1a** (5 mol%), Grignard reagent **3** (1.5 equiv., 1.0 M in THF), LiO^tBu (1.5 equiv.), THF (2.0 mL), 70 °C, 24 h

^b In brackets the selectivity for the reaction in the absence of LiO^tBu is given. As side product corresponding biphenyls were observed

^c Determined by GC-MS and ¹H NMR. In brackets the yield for the reaction in the absence of LiO^tBu is given

^d 47 % (22 %) cross coupling at the C-Br was observed

^e 28 % cross coupling at the C-CN was observed for the LiO^tBu free system

^f 2.5 equiv. **3** and LiO^tBu

^g 2.5 equiv. **3**, the isolated yield corresponds to the conversion of both substituents on the sulfur

observed for the methyl substituent in *para*-position, while in case of bromo- and cyano-substitution the cross coupling appeared at the bromo- or cyano-position (Table 2, entries 2–4). On the other hand 2-chloroethyl-phenylsulfide (**8**) gave the coupling product **4**, while a cross coupling (sp²–sp³) at the chloro-substituent was not observed (Table 2, entries 5 and 6). The reaction outcome was increased by addition of 2.5 equiv. of the Grignard reagent. Next, thioethers containing two sp²-carbons connected to the sulfur were tested (Table 2, entries 7–10). Diphenylsulfone (**9**) was converted in 73 % to the desired product. In contrast, for dibenzothiophene (**10**) as major product the 2,2'-disubstituted biphenyl **10a** was observed. Furthermore, the reaction with 2.5 equiv. of the Grignard reagents **3** and LiO^tBu gave higher yield of the coupling product **10a**. In addition, sulfoxides with aromatic substituents were reacted with the Grignard reagent **3** in the presence of **1a** (Table 2, entries 10–15). Good yields of the desired product were achieved, while for the benzyl sulfoxide (**14**) only a moderate amount of product was formed (Table 2, entry 16). Moreover, a comparable yield to phenyl sulfoxide (**12**) was observed for the conversion of the sulfone **15** (Table 2, entry 17 and 18). In addition, for symmetrical substrates (e.g., **9**, **12**, **13**, and **15**) 2.5 equivalents of the Grignard reagent **3** were used (Table 2, entries 8, 13, 15 and 18). Interestingly, for substrates **9** and **15**

both substituents were converted to the corresponding biaryl product.

In order to study the influence of the nucleophiles different Grignard reagents were reacted with thioanisole (**2**) (Table 3). Excellent yield was observed for the reaction with *p*-tolylmagnesium bromide (Table 3, entry 2). Furthermore, a variety of Grignard reagents were pre-generated by direct Mg insertion to the bromide in the presence of LiCl [62–69]. The cross coupling reaction of each Grignard reagents and thioanisole (**2**) in the presence of 5 mol% of pre-catalyst **1a** gave the desired coupling products in good to excellent yields (Table 3, entries 3–8). Moreover, alkyl Grignard reagents were examined, however resulting in no formation of product.

Moreover, for studying the abilities of the thioether function in comparison to other leaving groups an equimolar mixture of 4-iodoanisole (**23**), 4-bromoanisole (**24**), 4-chloroanisole (**25**), 4-fluoroanisole (**26**), and 4-methoxythioanisole (**27**) were reacted with stepwise increasing amounts of *p*-tolylmagnesium bromide (**16**, 1–3 equiv.) (Figure 1). After addition of the first equivalent of **16** as expected a decrease of the concentration of 4-iodoanisole (**23**) was noticed. Increasing the amount of equivalents of Grignard reagent the conversion of the 4-bromoanisole (**24**) and 4-methoxythioanisole (**27**) was observed, while

Table 3 Nickel-catalyzed cross coupling of thioanisole (**2**) and Grignard reagents—scope and limitations

	2	3,16-22		4,16a,18a-22a
Entry	Substrate	Product	Sel. [%] ^a	Yield [%] ^b
1	3		94 (>99)	4 : 77 (>99) ^c
2	16		93 (>99)	16a : 62 (78) ^c
3	17		>99	4 : 76 ^d
4	18		>99	18a : 73 ^d
5	19		>99	19a : 76 ^d
6	20		>99	20a : 64 ^d
7	21		>99	21a : 70 ^d
8	22		>99	22a : 95 ^d

^a Determined by GC-MS and ¹H NMR. In brackets the yield for the reaction in the presence of 1.5 equiv. LiO^tBu is given

^b In brackets theselectivity for the reaction in the absence of LiO^tBu is given. As side product corresponding biphenyls were observed

^c Reaction conditions: substrate (1.0 mmol), pre-catalyst **1a** (5 mol%), Grignard reagent (1.5 equiv., 1.0–2.0 M in THF), THF (2.0 mL), 70 °C, 24 h

^d Grignard reagents were pre-generated, see Ref. [62–69]. Reaction condition: substrate (0.7 mmol), aryl bromide (1.0 mmol), Mg (1.1 mmol), LiCl (1.0 mmol), pre-catalyst **1a** (5 mol%), THF (5 mL), 70°C, 24 h

4-chloroanisole (**25**) and 4-fluoroanisole (**26**) were almost unreacted. In more detail, the found reactivity is in line with the bond dissociation energy of each bonds (Ar–I: 64 kcal mol^{−1}, Ar–Br: 79 kcal mol^{−1}, Ar–Cl: 95 kcal mol^{−1}, Ar–F: 124 kcal mol^{−1}, Ar–SMe: 77 kcal mol^{−1}) [70].

Based on the preliminary results we propose a reaction mechanism as presented in Scheme 2 [52–56]. Initially the triphenylphosphane ligand dissociates from complex **1a** to create an open coordination site. In an earlier NMR study the dissociation was followed by ³¹P NMR and free coordination sites were occupied by the reaction solvent THF to

form the paramagnetic complex **1a-1** [52–54]. The complex **1a-1** reacts with the Grignard reagents to create complex **1a-2** or **1a-3** containing a 4-membered ring system [61]. The organic sulfide approaches the coordination sphere of the nickel and mediates a single electron transfer (SET) to oxidize the complex and produce a radical. The one-electron oxidized complex can be on the one hand oxidized at the metal center (**1a-4** or **1a-5**) or on the other hand the electron can be stored in the ligand (non-inocent ligand: **1a-6** or **1a-7**). Subsequently, RSMgX is eliminated and the radical recombines with the nickel species to form

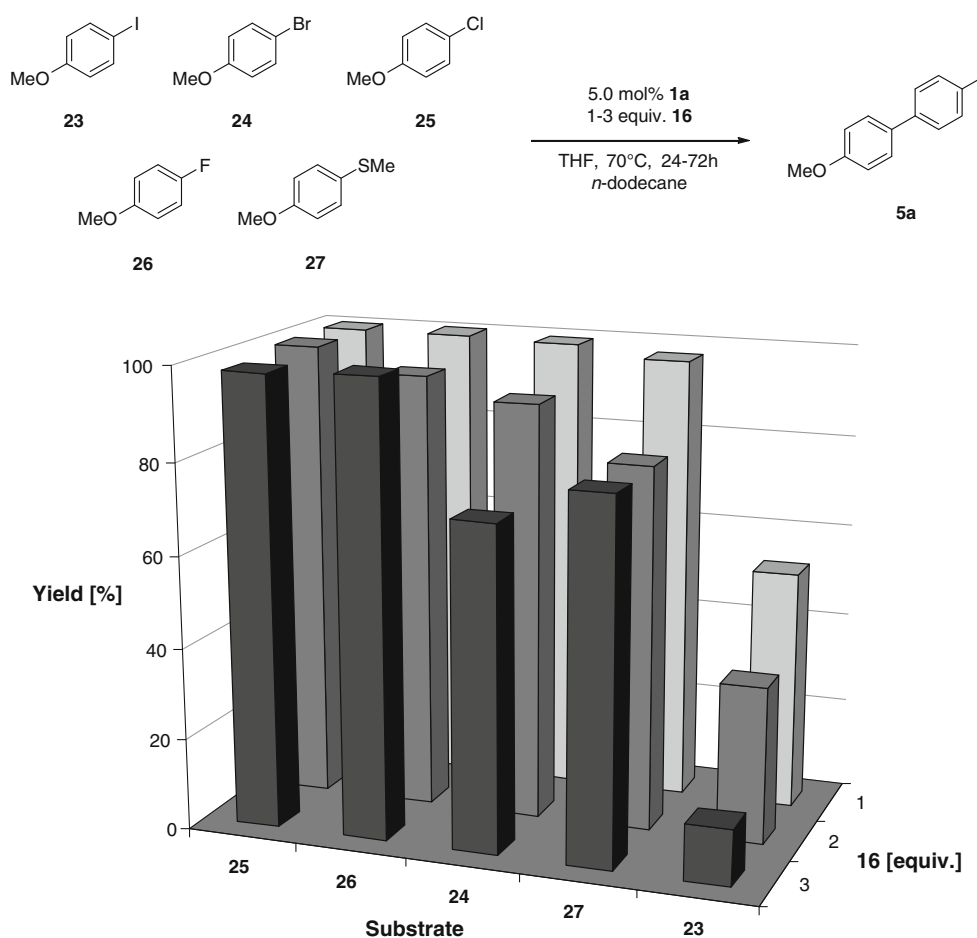
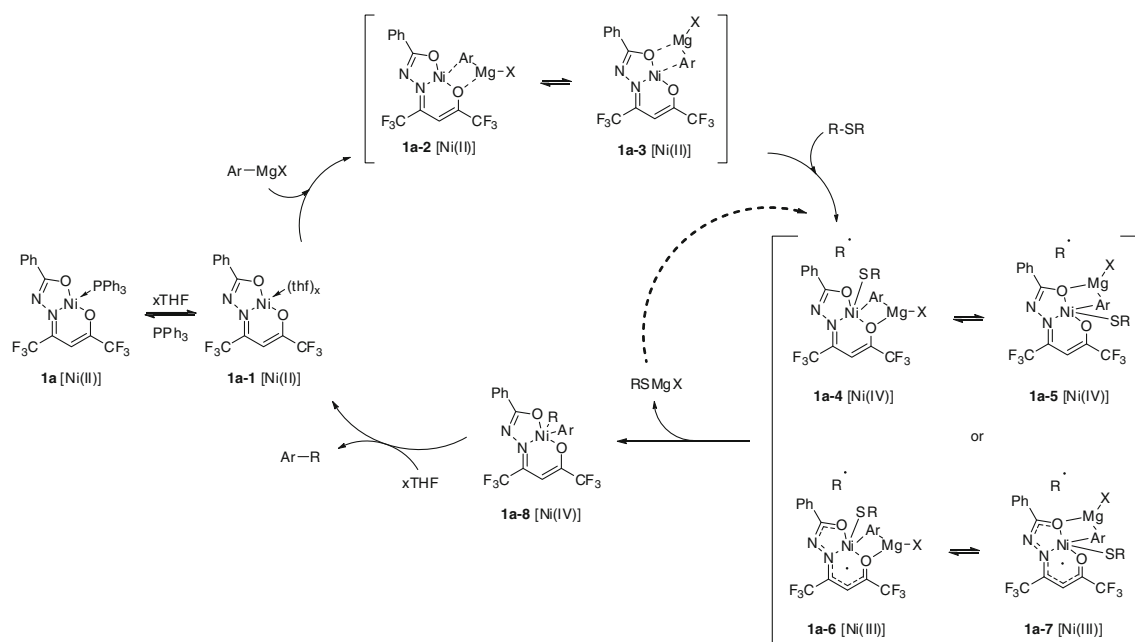


Fig. 1 Reactivity study. [Reaction conditions: **23**, **24**, **25**, **26**, and **27** (each 1.0 mmol), pre-catalyst **1a** (5 mol%, 0.05 mmol), **16** (every 24 h 1.0 equiv., 1.0 M in THF), THF (10 mL), 70 °C.]



Scheme 2 Proposed catalytic cycle for the nickel-catalyzed cross-coupling reaction of aryl Grignard reagents and organic sulfur compounds

the intermediate **1a-8**. Finally, an elimination of Ar–R occurs to regenerate complex **1a-1**.

In summary, we have set up a nickel-catalyzed carbon–carbon cross coupling protocol applying sulfur containing substrates as electrophiles and Grignard reagents as nucleophiles. With the well-defined nickel complex **1a** a variety of thioethers, sulfoxides and sulfones were converted to the desired products.

3 Experimental Section

3.1. General: All manipulations with oxygen- and moisture-sensitive compounds were performed under dinitrogen using standard Schlenk techniques. ^1H and ^{13}C were recorded on a Bruker AFM 200 spectrometer (^1H : 200.13 MHz; ^{13}C : 50.32 MHz) using the proton signals of the deuterated solvents as reference. GC–MS measurements were carried out on a Shimadzu GC-2010 gas chromatograph (30 m Rxi-5 ms column) linked with a Shimadzu GCMA-QP 2010 Plus mass spectrometer.

3.2. General procedure for the cross coupling reaction: A Schlenk flask was charged with an appropriate amount of complex **1a** (0.02 mmol, 5.0 mol%) and the corresponding thioether (0.3 mmol). The flask was cycled with nitrogen and vacuum. Afterwards THF (2.0 mL) and a THF solution of the corresponding Grignard reagent (1.5 equiv., 1.0 M in THF) were added. The flask was sealed and heated at 70 °C for 24 h or stirred at room temperature. After that time, the mixture was cooled, and dichloromethane and an aqueous NH_4Cl solution were added. The aqueous phase was extracted with dichloromethane and dried over Na_2SO_4 . *n*-Dodecane (internal standard) was added and an aliquote was taken and for GC–MS analysis. The solvent was removed and the residue was purified by column chromatography (eluent: toluene). The products were confirmed by GC–MS and NMR analysis. The analytical properties of the products are in agreement with literature data.

3.3. General procedure for the preparation of Grignard reagents via direct Mg insertion in the presence of LiCl: A 25 mL Schlenk tube flushed with nitrogen was charged with LiCl (1 mmol), Mg (1.1 mmol) and of THF (1.5 mL) was added under nitrogen. To the slurry solution of aryl bromide in THF (1 mmol was dissolved in 2.0 mL) was added and stirred vigorously. The reaction mixture was stirred over night at room temperature.

3.3.1. 4-Methoxy-1,1'-biphenyl [71] (4): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.68–7.75 (m, 4H), 7.61–7.47 (m, 3H), 7.13 (d, J = 8.7 Hz, 2H), 3.93 (s, 3H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 159.2, 140.8, 133.7, 128.8 (2C), 128.1 (2C), 126.6, 126.7 (2C), 114.3 (2C), 55.2 ppm. MS (EI) m/z = 185 (16), 184 (100, M^+), 169 (52), 141 (58), 139 (14), 115 (49), 63 (12), 40 (56).

3.3.2. 4-Methoxy-4'-methyl-1,1'-biphenyl [72] (5a): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.54–7.44 (m, 4H), 7.24–7.20 (m, 2H), 6.98–6.94 (m, 2H), 3.82 (s, 3H), 2.38 (s, 3H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 158.8, 138.0, 136.4, 133.8, 129.5 (2C), 128.0 (2C), 126.6 (2C), 114.3 (2C), 55.4, 21.1 ppm. MS (EI) m/z = 199 (15), 198 (100, M^+), 183 (50), 155 (32), 153 (11), 152 (11), 128 (12), 115 (11), 40 (16).

3.3.3. 4-Cyano-1,1'-biphenyl [73] (7a): MS (EI) m/z = 209 (25, M^+), 166 (11), 40 (100).

3.3.4. 2,2'-Bis(*p*-anisyl)-1,1'-biphenyl (10a): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.53–6.96 (m, 14H), 3.75 (s, 3H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 158.1 (2C), 140.7 (2C), 133.8 (2C), 131.7 (2C), 130.3 (3C), 130.0 (2C), 127.5 (2C), 126.9 (2C), 113.0 (3C), 55.3 (2C) ppm. MS (EI) m/z = 367 (28), 366 (100, M^+), 215 (12), 40 (50).

3.3.5. 1-Benzyl-4-methoxybenzene [74] (14a): MS (EI) m/z = 198 (27, M^+), 197 (10), 167 (11), 40 (100).

3.3.6. 4-Methyl-1,1'-biphenyl [75] (16a): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.71–7.33 (m, 8H), 2.50 (s, 6H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 141.3, 138.4, 136.8, 137.1, 129.6 (2C), 128.8 (2C), 127.2 (2C), 127.1 (2C), 21.2 ppm. MS (EI) m/z = 168 (51, M^+), 167 (33), 165 (13), 153 (10), 152 (13), 40 (100).

3.3.7. 4-(*tert*-Butyl)-1,1'-biphenyl [76] (18a): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.60–7.28 (m, 9H), 1.35 (s, 9H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 150.4, 141.2, 138.3, 128.8, 127.1, 126.9, 126.8, 125.8, 34.6, 31.5 ppm. MS (EI) m/z = 195 (100), 40 (67), 210 (37, M^+), 167 (30), 41 (20), 84 (17), 196 (17), 152 (13), 155 (11), 165 (11).

3.3.8. 4-(*N,N*-Dimethylamino)-1,1'-biphenyl [77] (19a): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.65–6.84 (m, 9H), 3.03 (s, 6H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 150.4, 141.2, 138.3, 128.8 (2C), 127.1 (2C), 126.9 (2C), 126.8 (2C), 125.8 (2C), 34.6, 31.5 (2C) ppm. MS (EI) m/z = 197 (100, M^+), 196 (73), 40 (53), 152 (21), 98 (19), 181 (17), 198 (16), 153 (12), 90 (12).

3.3.8. 2-Phenylnaphthalene [76] (20a): ^1H NMR (CDCl_3 , 200 MHz) δ = 8.03–7.33 (m, 12H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 141.3, 138.6, 133.9, 132.8, 129.0, 128.4, 128.3, 127.6, 127.5, 127.4, 126.4, 126.0, 125.9, 125.7 ppm. MS (EI) m/z = 204 (100, M^+), 202 (49), 40 (43), 203 (38), 101 (30), 205 (23), 102 (16), 89 (14), 76 (12), 88 (12).

3.3.9. 2,4,6-Trimethyl-1,1'-biphenyl [78] (21a): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.57–7.35 (m, 5H), 7.14 (s, 2H), 7.53 (s, 3H), 2.21 (s, 6H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 141.2, 139.2, 136.6, 136.0, 129.4, 128.4, 128.1, 126.6, 20.9, 20.8 ppm. MS (EI) m/z = 40 (100), 196 (30, M^+), 181 (25), 165 (14).

3.3.10. 3,5-Di-*tert*-butyl-1,1'-biphenyl [79] (22a): ^1H NMR (CDCl_3 , 200 MHz) δ = 7.82–7.41 (m, 8H), 1.59 (s, 18H) ppm. $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 50 MHz) δ = 151.2 (2C), 140.9, 142.7, 128.8 (2C), 127.6 (2C), 127.1, 121.9

(2C), 121.5, 35.11 (2C), 31.7 (6C) ppm. MS (EI) m/z = 40 (100), 251 (83), 57 (63), 266 (30, M^+), 41 (28), 252 (18).

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